Psychological Characteristics of Patients Undergoing Awake Craniotomy With Dexmedetomidine: A Retrospective Analysis

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BACKGROUND AND OBJECTIVES: Only few studies have investigated the psychological profile of patient candidates for awake craniotomy (AC). The aim of this study was to describe the multidisciplinary clinical protocol adopted in our hospital and to retrospectively explore differences in psychological characteristic of 51 patients, analyzing association with the tolerability of the procedure.

METHODS: During the preoperative neuropsychological assessment, the Pain Anxiety Symptoms Scale was administered to assess pain-related anxiety. Clinical characteristics and anesthesiological management were retrospectively recollected. The AC procedure was classified as "optimal" or "suboptimal" based on administration of rescue sedative midazolam doses because of psychomotor agitation or emotional distress in excess with respect to our anesthesiological protocol.

RESULTS: The awake procedure was considered optimal in 33 cases (64.7% of total sample). General anesthesia was never required in our patients. No significant differences between optimal and suboptimal groups were found. No single variable was significant as a predictor of outcome on the univariate analysis, and no multivariate predictive model was possible.

CONCLUSION: Although in our experience the Pain Anxiety Symptoms Scale score was not useful in identifying candidates for AC procedures, our multidisciplinary approach provided good results with respect to the tolerability of AC surgery. However, more studies are required to identify psychological profiles and preoperative possible warning signs to find the best course of surgery for each patient.

KEY WORDS: Awake mapping, Brain tumor, Dexmedetomidine, Patient selection, Psychological assessment

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Awake craniotomy (AC) is a gold standard for neurosurgical interventions that require tissue resection close to "eloquent" brain regions[1,2] to maximize tumor removal and prevent postoperative neurological deficits. This provides a better prognosis and improves the patient's quality of life.[3-5] During AC, the patient is asked to be responsive, allowing intraoperative cognitive monitoring and cortical mapping through a direct electrical stimulation.[2]

ABBREVIATIONS: AC, awake craniotomy; **MAC,** monitored anesthesia care; **MoCA,** Montreal Cognitive Assessment; **PASS-20,** Pain Anxiety Symptoms Scale.

Several studies reported that AC is generally well-tolerated with high level of patient's satisfaction.[6-8] Nevertheless, AC may expose patients to significant psychological stress with negative long-term consequences on mental health.[9] The procedure is generally contraindicated in case of psychiatric disorders, emotional instability, and claustrophobia because these conditions are associated with a higher probability of intraoperative psychomotor agitation, intolerance, or distress.[2,10,11] Discomfort, pain, and anxiety are common reported factors that negatively affect cooperation during brain mapping and prompt higher sedative drug doses, thus increasing the risk of adverse events.[12-16] Only few studies have evaluated the psychological profile of patient candidates for AC. Santini et al[10] investigated the possible correlation

between a preoperative psychological evaluation and patient's compliance during AC. These authors suggested that the structured interview "Pain Anxiety Symptoms Scale" (PASS-20) could play a role as a good predictor of patients' intraoperative behavior and that fear of pain and anxiety can be considered warning signs for minor failure.[11,12] According with this evidence, we have routinely introduced the use of PASS-20 in the preoperative psychological assessment of AC patients.

The goal of our article was to describe the multidisciplinary clinical protocol adopted in our hospital for AC and to study the possible relationship between baseline patients' psychological characteristics and the tolerability of the AC procedure. To this purpose, the outcome of the awake procedure was classified as optimal or suboptimal. According to previous studies,[10,13] we assume that the preoperative psychological evaluation may be informative and predictive of the tolerability of the procedure. Albeit retrospective, our study provides information to the multidisciplinary team to improve the selection of candidates for AC and suggests possible hints for further research on this subject.

METHODS

Study Design

Our observational retrospective cohort study aims at describing the multidisciplinary procedure adopted in our hospital to select candidates for AC. Moreover, we report on preoperative variables possibly affecting the quality of the awake surgery procedure. The study was approved by the local Ethics Committee. Because no intervention or monitoring different from our current clinical practice was implemented, our local Ethics Committee granted a waiver of informed consent. The STROBE checklist for reporting observational studies has been implemented.

Participants

This study included 51 patients who underwent AC surgery at Foundation IRCCS Neurological Institute C. Besta from January 2017 to August 2023. According to our Hospital's routine, AC was proposed by the neurosurgeon for any patient undergoing tumor ablation in close proximity with cortical areas or subcortical tracts involved in motor or language functions ("eloquent" areas).[17-21] Psychiatric disorders, claustrophobia, acquired language deficits, and patient's refusal were considered exclusion criteria for the AC procedure.[10,11] In these cases, general anesthesia was scheduled and patients were not enrolled in the study.

Settings

Psychological and Neuropsychological Assessment

Patients underwent a preoperative multidisciplinary stepwise assessment that includes medical evaluation and a neuropsychological assessment, consisting of a cognitive screening and a psychological interview (see Figure).

The cognitive screening was meant to evaluate the patient baseline performance for comparison with the postoperative performance. The Montreal Cognitive Assessment (MoCA)[22] was used as a rapid and general screening tool to assesses different cognitive domains: attention, executive functions, memory, language, visuoconstructional skills, conceptual thinking, calculations, and orientation. The MoCA total score is 30, and the cut-off score to define cognitive impairment is ≤26. According to current normative data, the cut-off point to define impairment was put at ≤5th percentile of cognitive functions adjusted on age, sex, and education.[17,23] Because a wide variety of impairments can result from a brain lesion resection,[17,18] intraoperative patient-specific tests were selected according to fMRI reports, preoperative neuropsychological performance, tumor site, and functional networks involved in the resection.[19-21]

The psychological interview involved psychological history collection and preparation of the patient to give information about the potential risks, explain the procedure in details, support the patient, and address his/her concerns.[22] This allows to establish a trusting relationship with the patient which is associated with higher intraoperative compliance and patient's satisfaction.[2,8] The Italian version of the Pain Anxiety Symptoms Scale (PASS-20)[24] was administered. It is a 20-item tool that measures pain-related anxiety on a Likert 0–5 scale. It is articulated in subscales: fear of pain (PASS F); cognitive anxiety (PASS CA); escape-avoidance behavior (PASS E/A); and physiological symptoms of anxiety (PASS Phy). The total score ranges from 0 (no pain anxiety) to 100 (severe pain anxiety).

Anesthesiological Management

All patients underwent routine preoperative anesthesiological evaluation. Before surgery, peripheral vein cannulation was implemented as required and a urinary bladder catheter was put in place. Standard monitoring (lead II ECG, invasive arterial pressure monitoring, Bispectral Index, ETCO2) was applied throughout surgery. Dexmedetomidine continuous ev infusion was started with a target loading dose of 0.5–1 mcg/kg/h, aiming at a clinical level of sedation of 3–4 on the Observer's Assessment of Alertness/Sedation Scale.[25] Thereafter, the dexmedetomidine infusion rate was maintained at 0.4–0.8 mcg/kg/h during craniotomy and decreased to 0.1–0.3 mcg/kg/h 10 minutes before the neurophysiological mapping (targeting an Observer's Assessment of Alertness/Sedation Scale of 5).[26,27] Locoregional anesthesia was provided by both scalp nerve block and surgical wound infiltration with 0.75% levobupivacaine.[28] Additional analgesic remifentanil (infusion of 0.01-0.05 mcg/kg/min) or sedative (midazolam) drug ev administration was decided by the attending anesthesiologist.

Awake Craniotomy Procedure

Patients were placed in a comfortable position on the operating table. No pin head fixation was applied. Air blown blankets were used for temperature control. The surgical field was prepared so that the patient was always visible by the neuropsychologist, who provided support and administered the cognitive monitoring.[27]

Surgery always took place with a neuronavigation system (Stealth 7 o 8, Medtronic Inc), according to a preoperative imaging evaluation. An image-guided minimally invasive approach was performed, with a linear skin incision followed by a small craniotomy centered on the preplanned trajectory to the lesion.[29] After dural opening, a 4-contacts strip electrode (SpesMedica) is carefully introduced in the subdural space to record electrocorticography and to perform phase-reversal and direct cortical stimulation motor evoked potential.

Cortical and subcortical mapping were performed to identify language and motor eloquent sites. Language mapping was performed with a bipolar electrical stimulator (tips separate by 10 mm, biphasic, total

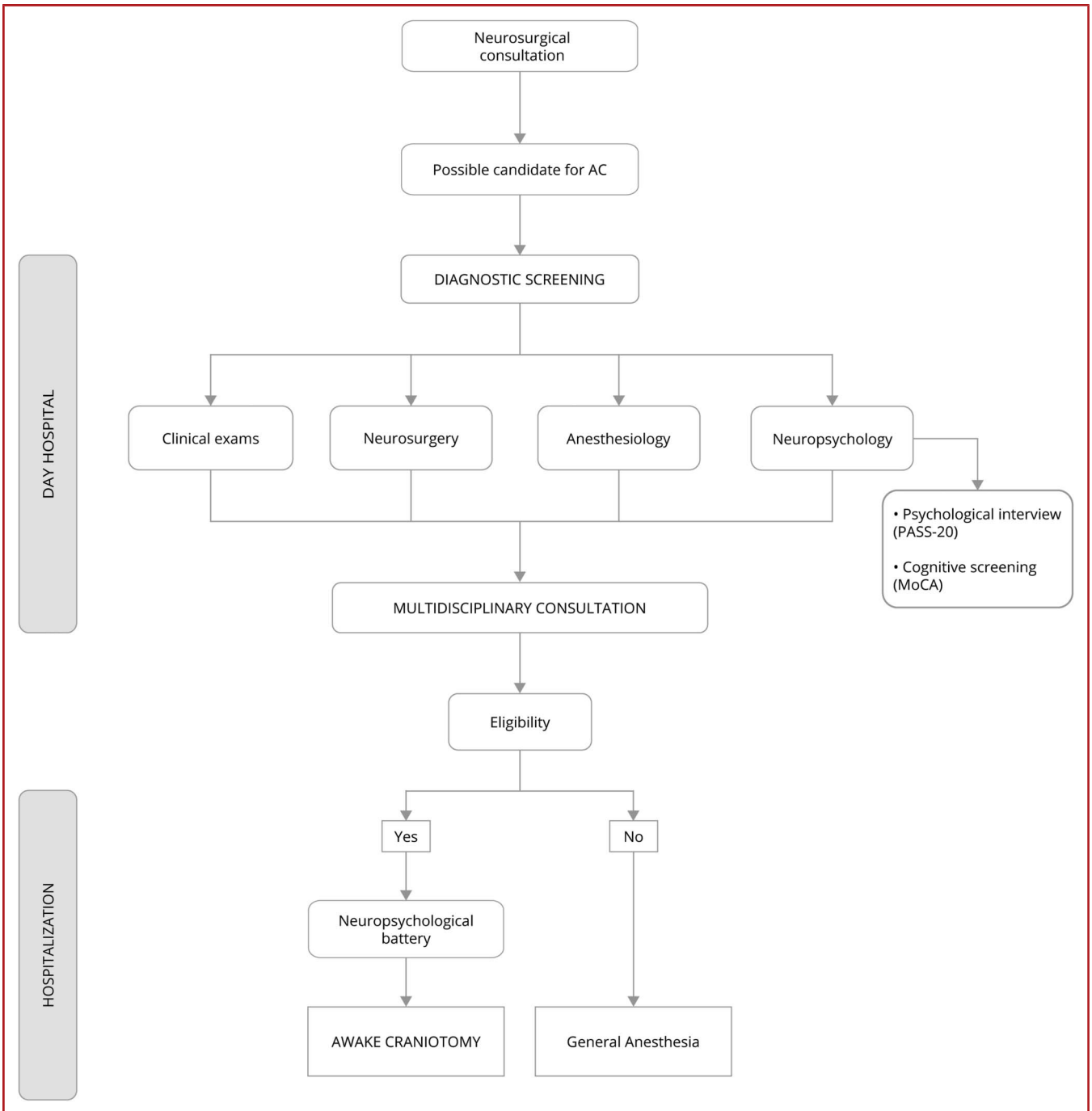


FIGURE. *Multidisciplinary step-wise assessment for AC candidates.*[16] *AC, awake craniotomy; MoCA, Montreal Cognitive Assessment; PASS-20, Pain Anxiety Symptoms Scale.*

duration 1 ms, 50 Hz, for 1-4 seconds) beginning at 2 mA and gradually increasing by 0.5 mA (8 mA maximum threshold in our series).[20,21,30] The ventral premotor cortex was firstly stimulated to obtain speech arrest during counting or naming tasks, to identify the intensity of the relevant working current, defined as the minimal intensity (usually between 3.5 and 8 mA) that produced anarthria without triggering epileptiform activity (after-discharges) on the electrocorticography. This current is then applied to suspect language eloquent areas while administering appropriate cognitive tests (counting, object naming, verbal fluency, word repetition, verb generation, reading, orofacial praxis).[19-21] If stimulation

induced impairment of the tested function, it was defined positive. A site is considered eloquent if an error occurs on 3 nonconsecutive tests.[20,21]

Motor mapping was performed with a monopolar electrical stimulator (short-train technique), applying anodal stimulation for cortical mapping and cathodal stimulation for subcortical mapping.[31] Electromyography was required for MEP registration and allowed early recognition of muscle spread activity because of seizures.[32]

Variables

To qualify the successfulness of the awake surgery procedure, we defined an outcome binary variable that scored the AC procedure as "optimal" or "suboptimal." The AC procedure was defined "optimal" if it was accomplished with the patient always awake, collaborative, and well-controlled without any need of rescue maneuvers. It was meant "suboptimal" if it could be accomplished only after administration of rescue sedative midazolam doses because of psychomotor agitation or emotional distress. Had general anesthesia be required to accomplish neurosurgery, this would have been scored as "suboptimal." Clinical data, including tumor location, tumor dimensions (>4 cm), duration of surgery, length of hospitalization and anesthesiological management, were obtained from medical records.

Preoperative PASS scores and the MoCA cognitive score were studied in a logistic model as possible predictors of the outcome, while male sex, age (years), education (years of attendance), and duration of surgery (min) were taken as possible covariates.

Statistical Methods

Statistical analysis was performed using R Core Team (2020).[33] Continuous data are reported as mean ± SD [median (IQR)]. Categorical data are reported as number (percentage). PASS-20 scores were categorized according to validated cut-off of normative data (scores >1 SD and >2 SD).[34] AC successfulness (optimal vs suboptimal) was taken as the outcome variable. Potentially predictive variables and covariates were analyzed with logistic regression in a univariate variable selection taking $P < .20$ as the cut-off value. In a possibly resulting multivariate logistic predictive model, significance was considered for $P < .05$.

RESULTS

Descriptive Data

Table reports on the descriptive statistics of our series. Seventeen patients (33.3% of total sample) had lesions in frontal lobe, 13 patients (25.5%) in temporal areas, 4 patients (7.8%) in parietal lobe, and only 1 in insular regions (2.0%). Fifteen patients (29.4%) had brain lesions in large areas involving multiple lobes (60.8% of the total sample had lesions dimensions greater than 4 cm). Only 1 patient underwent AC for epilepsy (2.0%). Preoperative PASS-20 and MoCA scores were also reported (see Table).

Main Results

The awake procedure was considered optimal in 33 cases (64.7% of total sample). General anesthesia was never required in our patients. No significant differences between optimal and suboptimal groups were found. No single variable was significant as a predictor of outcome on the univariate analysis, and no multivariate predictive model was possible.

DISCUSSION

Key Results

The goal of our article was to describe the multidisciplinary clinical protocol adopted in our hospital for AC and to study the possible relationship between patients' psychological characteristics and the tolerability of the awake surgery procedure.

Several studies showed how pain and agitation may lead to detrimental effects, which compromise the success of the procedure by impairing patients' cognitive functions and cooperation.[12-16] It has been reported that preoperative fear and anxiety were correlated with pain perception.[9,10] However, only few studies have explored preoperative psychological conditions and their association with the surgical outcome, and inconsistent results were reported.

Therefore, we investigated whether preoperative pain-related anxiety evaluation might support clinical decision making in the selection of patient candidates for AC. Because PASS-20 has been indicated as a good predictor of intraoperative behavior, which can even help to determine patient eligibility to AC, in the past decade, we introduced it in the preoperative routine assessment of AC candidates.[10]

To qualify the successfulness of the AC procedure, an outcome binary variable scored the AC procedure as "optimal" or "suboptimal" based on the requirement of sedation. As an indicator of excessive psychomotor agitation or emotional distress, we considered the requirement for intraoperative anxiolytic medications during surgery. The anesthetic management is a critical component in this setting because it is meant to reduce patients' psychophysical distress, thus providing safety and effectiveness to the procedure.[14,27]

In contrast to the study of Santini et al,[10] our results showed no statistically significant difference between the optimal and the suboptimal group in the level of pain-related anxiety assessed with the PASS-20. In addition, none of the potentially predictive variables we studied was predictive of the outcome. Some methodological issues could account for this.

In the study of Santini et al,[10] an asleep-awake technique involving general anesthesia and maneuver on the airways was performed. In our series, a monitored anesthesia care (MAC) approach with dexmedetomidine was provided, allowing patients to breath spontaneously throughout the procedure.[35,36] Moreover, although both studies focused on patients' compliance, we selected a dichotomic outcome based on the need of rescue sedation, while Santini et al evaluated the patient's intraoperative behavior using a self-report interview assessing fear and pain. Although sensitive and informative, this patient-reported measure could also be less objective and more susceptible to interindividual variability. Moreover, in our study, the cognitive general screening assessed with the MoCA test (24.7 ± 3.3) did not provide any

TABLE. Demographic and Clinical Characteristics of Patients Who Underwent AC

Characteristics	Overall n = 51	Optimal 33 (64.7%)	Suboptimal 18 (35.3%)
Age (y)	43.1 ± 14.1 [44.0 (30.0-54.0)]	44.4 ± 14.8 [44.0 (30.0-55.0)]	40.6 ± 12.7 [40.5 (30.0-49.7)]
Education (y of attendance)	14.2 ± 4.0 [13.0 (13.0-18.0)]	14.2 ± 4.0 [13.0 (13.0-18.0)]	14.1 ± 4.1 [13.0 (13.0-17.7)]
Duration of surgery (min)	248.4 ± 77.6 [250.0 (200.0-300.0)]	241.1 ± 84.4 [245.0 (190.0-300.0)]	261.9 ± 63.4 [250.0 (240.0-300.0)]
Length of hospital stay (d)	5.0 ± 1.4 [5.0 (4.0-6.0)]	4.9 ± 1.2 [5.0 (4.0-6.0)]	5.3 ± 1.7 [5.0 (4.0-6.0)]
Dexmedetomidine ev infusion max (mcg/kg/h)	1.1 ± 0.4 [1.0 (0.80-1.33)]	1.0 ± 0.3 [1.0 (0.80-1.30)]	1.1 ± 0.4 [1.1 (1-1.40)]
Remifentanil ev administration (yes/no)	20 (39.2%)	9 (27.3%)	11 (61.1%)
Male sex	30 (58.8%)	21 (63.6%)	9 (50.0%)
Lesion size (>4 cm)	31 (60.8%)	19 (57.6%)	12 (66.6%)
PASS-20			
Total score (max score 100)	36.9 ± 10.4 [37.0 (29.8-41.0)]	37.2 ± 10.7 [37.0 (30.5-40.5)]	36.2 ± 10.0 [33.0 (30.0-41.0)]
Cognitive anxiety (max score 25)	10.6 ± 3.6 [10.0 (8.0-12.3)]	10.8 ± 4.0 [10.0 (8.0-13.0)]	10.1 ± 3.0 [10.0 (8.0-12.0)]
Escape and avoidance (max score 25)	10.7 ± 3.4 [11.0 (8.0-13.0)]	10.7 ± 3.5 [11.0 (8.5-13.0)]	10.8 ± 3.1 [11.0 (8.0-12.0)]
Fear appraisal (max score 25)	7.9 ± 2.9 [7.0 (6.0-10.0)]	8.1 ± 2.9 [7.0 (6.0-10.0)]	7.6 ± 3.1 [6.0 (6.0-8.0)]
Physiological anxiety (max score 25)	7.7 ± 2.8 [7.0 (5.0-9.0)]	7.6 ± 2.7 [7.0 (5.0-8.5)]	7.8 ± 3.1 [7.0 (6.0-9.0)]
MoCA (max score 30)	24.7 ± 3.3 [25.0 (22.5-27.0)]	24.2 ± 3.6 [25.0 (21.0-27.0)]	25.8 ± 2.2 [25.0 (25.0-27.5)]

AC, awake craniotomy; MoCA, Montreal Cognitive Assessment; PASS-20, Pain Anxiety Symptoms Scale.
For continuous variables mean ± SD [median (IQR)]. Total number and percentage are reported for categorical variables. For each variable, units and maximal scores are presented. Dexmedetomidine ev infusion max refers to maximal endovenous infusion rate achieved during AC. Remifentanil ev administration (yes/no) refers to number of cases where endovenous administration was performed.

useful information, although severe cognitive impairments are known to seriously compromise the patient's cognitive reserve and affect tolerance and compliance during AC procedures.[37,38]

Our preoperative psychological assessment was not informative with respect to the tolerability of AC. Nevertheless, we observed a high prevalence (64.7%) of optimal AC, which suggests that our multidisciplinary management of awake patients works well in maintaining a good level of tolerance and compliance throughout the procedure. It should be also noted that all our cases were accomplished with a MAC approach, and general anesthesia was never required. Therefore, our exclusion criteria for patient's selection can be considered satisfactory.

Limitations

Our study has some limitations. It presents all the known limitations of observational retrospective designs. Retrospective data collection through medical records could be incomplete or imprecise.

A major limitation of our study is the relatively small sample size, and a type II error could account for the lack of significance we observed. With respect to this, our results should be considered preliminary.

Moreover, we defined the difference between "optimal" and "suboptimal" outcomes only on operator-evaluated criteria, without taking into account patient-evaluated variables such as subjective feelings of distress. Such variables should be included in future studies on AC because the anesthesiologist could underestimate possible emotional distress of patient with a high positive attitude. It should be also noted that our criteria for scoring an AC procedure as "suboptimal" is based on the administration of midazolam in excess with respect to our anesthesiological protocol. Although such protocol is conceivably acceptable in the AC setting and reflects a consolidated clinical practice, different sedation protocols can yield different results. The external validity of our results is also limited to the dexmedetomidine MAC approach we implemented.

Finally, we focused on patients who actually underwent awake surgery. Patients excluded from awake procedures could also be studied with a similar psychological assessment to provide a useful comparison.

CONCLUSION

Although in our experience the PASS-20 score was not useful in identifying candidates for AC procedures, our multidisciplinary approach provided good results with respect to the tolerability of AC surgery.

Our data could serve as a basis for future prospective studies aiming at the definition of possible psychological profiles and preoperative warning signs to optimize AC procedures.

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The authors have no personal, financial, or institutional interest in any of the drugs, materials, or devices described in this article.

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